

Deployment Runbook of the Nkwen Traders Site

This document explains how the Nkwen Traders static site is hosted on Amazon S3, and how to redeploy it if I or a Cloud Engineer is unavailable. The nkwen-traders-beacon website was deployed using Amazon S3 as the origin and Amazon CloudFront as the content delivery network (CDN).

The deployment provides HTTPS access to the website and uses CloudFront caching to improve content delivery.

Hosting method and Bucket Configuration

I used the Amazon S3 static website hosting to host the nkwen-traders-beacon website live.

1. I fill in exact bucket name, i.e. nkwen-traders-beacon. Then I filled in the region i.e. eu-west-3.
2. Under “Properties → Static website hosting”, I enabled hosting with:
 - **Index document:** index.html
 - **Error document:** 404.html
3. Under “Permissions → Block public access (bucket settings)”, I unchecked the settings blocking public bucket policies. This step is required, because S3 rejects any public bucket policy while Block Public Access is on, and will fail to save the policy with a permissions-style error even when the policy JSON itself is valid.
4. If the account has account-level Block Public Access settings, those must also be unchecked, since they override the bucket-level setting.

Bucket policy

The following policy grants public **read-only** access to the site's files only and nothing broader:

JSON Policy:

```
{  
  "Version": "2012-10-17",  
  "Statement": [  
    {  
      "Effect": "Allow",  
      "Principal": "*",  
      "Action": "s3:GetObject",  
      "Resource": "arn:aws:s3:::nkwen-traders-beacon/*"
```

```
{
  "Sid": "PublicReadForWebsite",
  "Effect": "Allow",
  "Principal": "*",
  "Action": "s3:GetObject",
  "Resource": "arn:aws:s3:::nkwen-traders-beacon-122794302986-eu-west-3-an/*"
}
```

Uploading and Redeploying the Site

To deploy or update the live site, upload the full contents of the frontend/ folder to the **root** of the bucket, preserving the folder structure (css/, js/, assets/, and the HTML pages). To upload the file, go to the AWS Console: open the bucket → Upload → drag in the contents of frontend/.

Limitation of the S3 Website Endpoint (HTTP Only)

The plain S3 website endpoint serves traffic over “HTTP, not HTTPS”. This is a limitation of S3 static website hosting itself, not a misconfiguration. S3 does not support HTTPS on its native website endpoints. This was flagged to the Security Reviewer as part of the pre-launch review.

Adding HTTPS via CloudFront

To serve the site over HTTPS:

1. Create a CloudFront distribution with the S3 website endpoint (not the bucket ARN) as a custom origin.
2. CloudFront provides a free cloudfront.net HTTPS URL with an AWS-managed certificate, no custom domain required.
3. Allow 10–15 minutes for the distribution to deploy.
4. Once live, use the CloudFront URL as the original link rather than the raw S3 endpoint.

Deployment Test

The CloudFront distribution was deployed successfully and the website was accessed through the CloudFront HTTPS URL: <https://d3ohwdhyzctk8j.cloudfront.net>. The website loaded successfully, confirming that CloudFront was correctly serving the S3 website.

A change was made to the website content in S3 to test CloudFront caching. A CloudFront invalidation was then created using: `"/*`". The invalidation completed successfully, and the updated website content was verified through the CloudFront URL.

Cost Considerations

The CloudFront distribution uses the Pay-as-you-go pricing model. AWS resources should be monitored regularly to prevent unexpected charges.

Troubleshooting

- "If Your bucket policy changes can't be saved": check both bucket-level and account-level Block Public Access settings (Section 2, step 3–4) before assuming an IAM permissions issue.
- If your Page shows a 403 or 404 after upload: confirm the file was uploaded to the bucket root (not nested inside an extra frontend/ folder), and that the bucket policy is active.
- If the catalog page is not loading products: confirm products.json was uploaded to the correct path and that the path referenced in the JS fetch() call matches.

Conclusion

The Nkwen Traders Beacon website was successfully deployed to Amazon S3 and distributed through Amazon CloudFront. HTTPS redirection, caching, and cache invalidation were tested successfully, and the website is accessible through the CloudFront distribution URL.